

HVAC Controls & Building Management System Specifications

Doc No: HVAC-003

Rev: A

Date: 2026-05-12

Project: Lakefront Office Tower — Phase II

ACME Engineering Associates, Inc.
123 Industrial Pkwy, Suite 400
Chicago, IL 60601

SECTION 23 09 23 — DIRECT DIGITAL CONTROLS (DDC) FOR HVAC

1.0 SCOPE

Furnish, install, program, and commission a fully integrated Building Management System (BMS) providing Direct Digital Control (DDC) of all mechanical systems. The BMS shall interface with the electrical, fire alarm, security, and lighting control systems via BACnet/IP protocol.

2.0 SYSTEM ARCHITECTURE

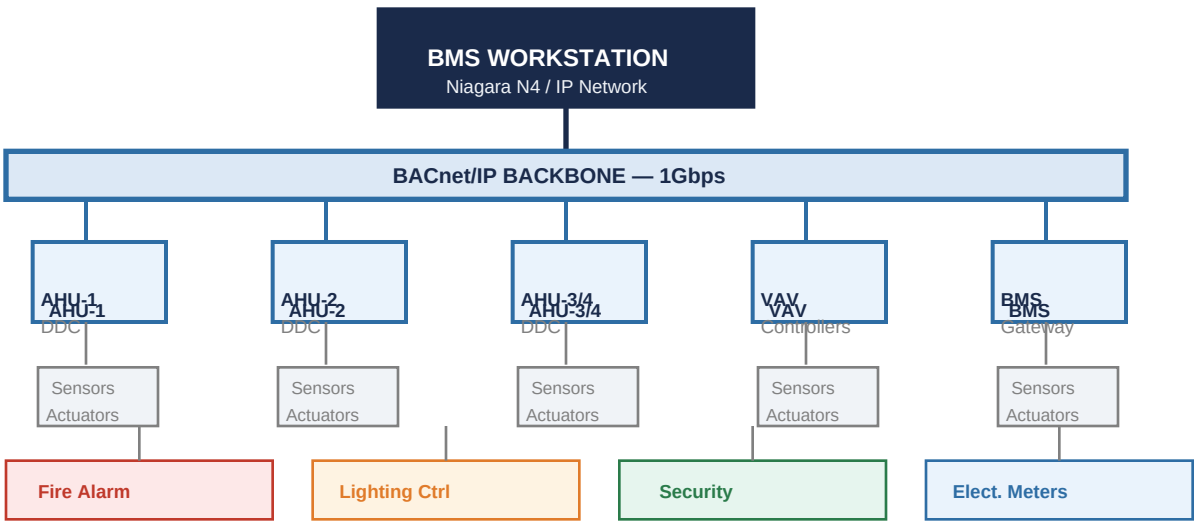


Figure 5 — BMS Network Architecture Diagram (Schematic — Not to Scale)

3.0 AHU CONTROL SEQUENCES

3.1 Supply Air Temperature Reset

SAT setpoint shall reset based on outdoor air temperature (OAT) as follows:

OAT (°F)	SAT Setpoint (°F)
≥ 75	55
65	57
55	60
≤ 45	65

3.2 Economizer Control

Differential enthalpy economizer shall modulate OA/RA/Exhaust dampers to maximize free cooling when OA enthalpy < RA enthalpy. High-limit shutoff: OA dry-bulb > 75°F or OA enthalpy > 28 BTU/lb. Comply with ASHRAE 90.1-2022 Section 6.5.1.

3.3 VAV System Pressure Reset (Static Pressure Reset)

Duct static pressure setpoint shall reset downward from 1.5 in. w.g. toward 0.5 in. w.g. based on VAV box demand. When all served VAV boxes can satisfy zone load with damper <90% open, reduce setpoint by 0.1 in. w.g. per 5-min interval. Per ASHRAE 90.1-2022 §6.5.3.2.

4.0 POINTS LIST — AHU-1

Point ID	Description	Type	Range / State
AHU1-MAT	Mixed Air Temp	AI	–20 to 120°F
AHU1-SAT	Supply Air Temp	AI	40 to 120°F
AHU1-RAT	Return Air Temp	AI	40 to 120°F
AHU1-OAT	Outdoor Air Temp	AI	–30 to 130°F
AHU1-SAP	Duct Static Pressure	AI	0 to 5 in. w.g.
AHU1-SAFLOW	Supply Airflow (Pitot)	AI	0 to 40,000 CFM
AHU1-CHWV	CHW Valve Position	AO	0–100%
AHU1-HHWV	HHW Valve Position	AO	0–100%
AHU1-OADMP	OA Damper	AO	0–100%
AHU1-EADMP	Exhaust Air Damper	AO	0–100%
AHU1-SFVFD	Supply Fan VFD Speed	AO	0–100% (min 15%)
AHU1-SFSS	Supply Fan Start/Stop	DO	Off/On
AHU1-SFST	Supply Fan Status	DI	Off/On
AHU1-SFFL	Supply Fan VFD Fault	DI	Normal/Fault
AHU1-FLTST	Filter Δ P Status	DI	Normal/Dirty
AHU1-FZDTK	Freeze-Stat Alarm	DI	Normal/Trip

END OF SECTION 23 09 23